



TRADITIONAL FILE PROCESSING SYSTEMS

When computer-based data processing was first available, there were no databases. To be useful for business applications, computers had to store, manipulate, and retrieve large files of data. Computer file processing systems were developed for this purpose. Although these systems have evolved over time, their basic structure and purpose have changed little over several decades.

As business applications became more complex, it became evident that traditional file processing systems had a number of shortcomings and limitations (described next). As a result, these systems have been replaced by database processing systems in most business applications today. Nevertheless, you should have at least some familiarity with file processing systems since understanding the problems and limitations inherent in file processing systems can help you avoid these same problems when designing database systems. It should be noted that Excel files, in general, fall into the same category as file systems and suffer from the same drawbacks listed below. Informal use of Excel for management of data is believed to continue to be quite widespread, although valid research results regarding this are difficult to find.

File Processing Systems at Pine Valley Furniture Company



Early computer applications at Pine Valley Furniture used the traditional file processing approach. This approach to information systems design met the data processing needs of individual departments rather than the overall information needs of the organization. The information systems group typically responded to users' requests for new systems by developing (or acquiring) new computer programs for individual applications, such as inventory control, accounts receivable, or human resource management. No overall map, plan, or model guided application growth.

Three of the computer applications based on the file processing approach are shown in Figure 1-2. The systems illustrated are Order Filling, Invoicing, and Payroll.

FIGURE 1-2 Old file processing systems at Pine Valley Furniture Company

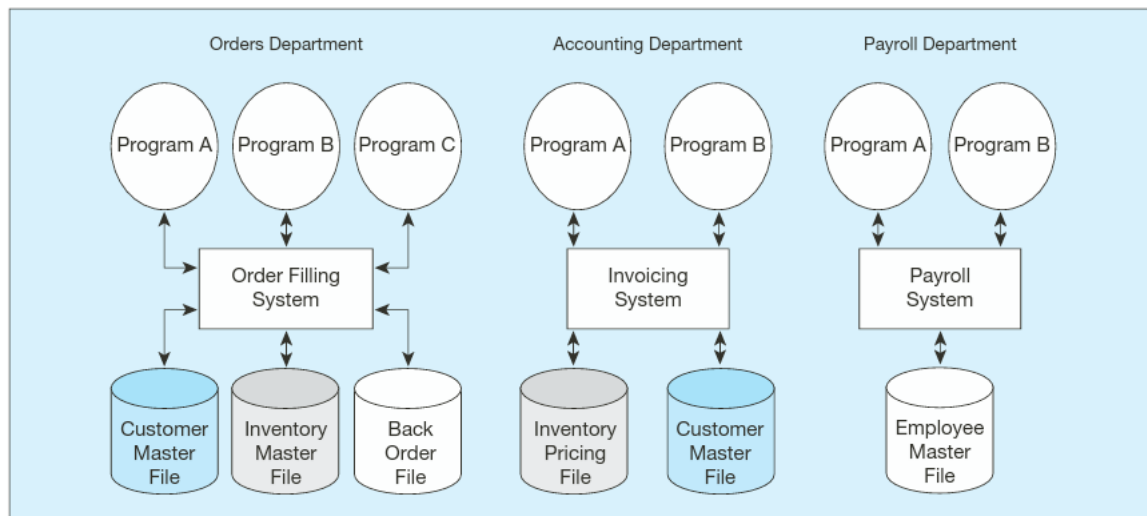



TABLE 1-2 Disadvantages of File Processing Systems

Program-data dependence
Duplication of data
Limited data sharing
Lengthy development times
Excessive program maintenance

The figure also shows the major data files associated with each application. A *file* is a collection of related records. For example, the Order Filling System has three files: Customer Master, Inventory Master, and Back Order. Notice that there is duplication of some of the files used by the three applications, which is typical of file processing systems.

Disadvantages of File Processing Systems

Several disadvantages associated with conventional file processing systems are listed in Table 1-2 and described briefly next. It is important to understand these issues because if you do not follow the database management practices described in this book, some of these disadvantages can also become issues for databases as well.

Database application

An application program (or set of related programs) that is used to perform a series of database activities (create, read, update, and delete) on behalf of database users.

PROGRAM-DATA DEPENDENCE File descriptions are stored within each **database application** program that accesses a given file. For example, in the Invoicing System in Figure 1-2, Program A accesses the Inventory Pricing File and the Customer Master File. Because the program contains a detailed file description for these files, any change to a file structure requires changes to the file descriptions for all programs that access the file.

Notice in Figure 1-2 that the Customer Master File is used in the Order Filling System and the Invoicing System. Suppose it is decided to change the customer address field length in the records in this file from 30 to 40 characters. The file descriptions in each program that is affected (up to five programs) would have to be modified. It is often difficult even to locate all programs affected by such changes. Worse, errors are often introduced when making such changes.

DUPPLICATION OF DATA Because applications are often developed independently in file processing systems, unplanned duplicate data files are the rule rather than the exception. For example, in Figure 1-2, the Order Filling System contains an Inventory Master File, whereas the Invoicing System contains an Inventory Pricing File. These files contain data describing Pine Valley Furniture Company's products, such as product description, unit price, and quantity on hand. This duplication is wasteful because it requires additional storage space and increased effort to keep all files up to date. Data formats may be inconsistent, data values may not agree, or both. Reliable metadata are very difficult to establish in file processing systems. For example, the same data item may have different names in different files, or, conversely, the same name may be used for different data items in different files.

LIMITED DATA SHARING With the traditional file processing approach, each application has its own private files, and users have little opportunity to share data outside their own applications. Notice in Figure 1-2, for example, that users in the Accounting Department have access to the Invoicing System and its files, but they probably do not have access to the Order Filling System or to the Payroll System and their files. Managers often find that a requested report requires a major programming effort because data must be drawn from several incompatible files in separate systems. When different organizational units own these different files, additional management barriers must be overcome.

LENGTHY DEVELOPMENT TIMES With traditional file processing systems, each new application requires that the developer essentially start from scratch by designing



new file formats and descriptions and then writing the file access logic for each new program. The lengthy development times required are inconsistent with today's fast-paced business environment, in which time to market (or time to production for an information system) is a key business success factor.

EXCESSIVE PROGRAM MAINTENANCE The preceding factors all combined to create a heavy program maintenance load in organizations that relied on traditional file processing systems. In fact, as much as 80 percent of the total information system's development budget might be devoted to program maintenance in such organizations. This, in turn, means that resources (time, people, and money) are not being spent on developing new applications.

As discussed above, these disadvantages are true also in situations when individuals and organizational units maintain important organizational data in Excel spreadsheets. Further, it is important to note that many of the disadvantages of file processing you have learned about can also be limitations of databases if an organization does not properly apply the database approach. For example, if an organization develops many separately managed databases (say, one for each division or business function) with little or no coordination of the metadata, uncontrolled data duplication, limited data sharing, lengthy development time, and excessive program maintenance can occur. Thus, the database approach, which is explained in the next section, is as much a way to manage organizational data as it is a set of technologies for defining, creating, maintaining, and using these data.